A Stochastic Model for Generating Synthetic Daily Household Electricity Demand Profiles from Smart-Meter Data – Validation on the Optimal Sizing of Batteries

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A data-driven stochastic model is proposed to generate synthetic daily household electricity demand scenarios for a distribution grid. The method clusters households by their average demand profiles to identify distinct consumption patterns and fits a multivariate Gaussian mixture model (GMM) to the logarithm of the demand residuals relative to each cluster mean. This formulation accurately captures both the average and median demand profiles of the original dataset across different days of the week and periods of the year. The model's ability to reproduce key statistical properties of the original data is evaluated with various key performance metrics, with particular emphasis on the joint distribution of daily peak power and energy consumption, which governs the coupling between the grid's power and energy needs. To demonstrate the practical value of the method, an optimal siting and sizing problem for a battery energy storage system (BESS) in a distribution network is formulated, and the results obtained using the synthetic scenarios are compared against average demand profiles reported in the literature. Because the peak-power-to-energy ratio is accurately modeled, the sizing algorithm computes 3.5 times less energy capacity for the considered use case, requiring 40% less investment to meet the same power requirements. The proposed method is of direct interest to distribution system operators (DSOs) for evaluating both wire and non-wire alternatives for network reinforcement.

CCS Concepts: • **Mathematics of computing → Multivariate statistics**; *Time series analysis*; *Maximum likelihood estimation*; • **Computing methodologies → Modeling methodologies**.

Additional Key Words and Phrases: Gaussian mixture models, load consumption, optimal sizing of batteries, peak-power-to-energy ratio, scenarios generation, statistical models

1 Introduction

Electrification and energy transition efforts require improved management and decision-support methods for distribution grids. Highly spatio-temporally resolved modeling and data on distributed energy resources (DERs) and loads are critical for evaluating the hosting capacity of existing distribution grids and for planning both wire-based and non-wire grid reinforcement measures. For example, evaluating grid upgrades is typically performed using methods such as optimal power flow or Monte Carlo simulations (e.g., [1–3]), which require modeling uncertain generation and load profiles as inputs to the problem.

Power flow studies primarily rely on average demand profiles. However, these may underestimate reinforcement needs because they can neglect the high-power demand peaks that characterize individual households [4]. When sizing battery energy storage systems (BESS) as an alternative to grid reinforcement, special care must be devoted not only to peak demand but also to total electricity consumption, as this directly impacts energy storage requirements. In this context, accurate modeling of the relationship between peak power and energy consumption is required to properly capture the coupling between the grid's power and energy needs.

It is also worth noting that although smart meter data are becoming increasingly available through the progressive rollout of smart meters in many countries and can support this type of analysis, such data may not be universally accessible or even available at all due to a lack of real-time IT infrastructure in many distribution system operators' (DSOs) networks. Moreover, their temporal resolution may be insufficient to characterize short-duration demand peaks, as smart meter measurements are typically collected at 15-minute intervals. In addition, for planning studies, measurements are naturally unavailable. In this context, models capable of generating synthetic data remain highly valuable for supporting decision-making.

In this context, this paper tackles the problem of generating realistic residential load data with high spatial disaggregation and high temporal resolution to support distribution grid analysis and planning. We propose a data-driven method for generating synthetic daily household demand scenarios for a distribution grid across different days and periods of the year.

Methods for modeling household consumption in the literature are typically classified as top-down or bottom-up (e.g., [5, 6]). Traditional top-down methods use historical aggregate demand data, together with macroeconomic and other exogenous variables, to model average electricity demand for a large area (e.g., [7]). Although these techniques can efficiently forecast consumption patterns for large-area studies, they fail to model the intra-day consumption of individual users.

Most recent methods follow a bottom-up approach, modeling household energy consumption using data on user behavior or even at the appliance level. Bottom-up models can be further classified into statistical models and engineering models. Statistical models train a stochastic model (e.g., [8–10]) using a combination of user-behavior data from time-of-use surveys (TOUS) and weather data to estimate individual users' energy consumption. On the other hand, engineering methods use physical models of appliances (e.g., [11]) to estimate household demand. Modeling at the device level enables high-resolution (even 1-minute) computation of demand scenarios that can efficiently capture peak demands during the day. However, these approaches require large amounts of data, which are not always available due to privacy concerns. A combination of top-down and bottom-up approaches was used in [12] by integrating historical measurements and user-behavior data, but the study estimated demand at the city level.

Recent studies have focused on data-driven, stochastic methods that learn consumption patterns from historical measurements of household demand without the need for explicit modeling of user or appliance behavior. These methods become appealing as large datasets of smart-meter measurements are becoming available (e.g.,

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[13, 14]). Smart-meter data are more likely to be available than TOUS data as they meet privacy requirements.

In this context, the method in [15] groups houses using mixture-model clustering to identify consumption patterns, and then trains a Markov decision process by fitting a Weibull distribution to the power histogram to generate daily load profiles. The model accurately captured the mean demand and distribution of daily energy consumption. However, the peak-power-to-energy ratio was not validated. [16] follows a similar approach but uses hierarchical clustering and fits a Gaussian Mixture Model (GMM) to the power distribution at each time step, which can model more complex distributions.

[17] uses a purely data-driven stochastic framework to model electricity demand at high resolution (sub-minute), capturing the high intermittency of demand in low-voltage grids. For this purpose, it employs empirical mode decomposition to extract average load profiles and superstatistics to model fluctuations about the trend. Similarly, [18] decomposes high-resolution data into a low-frequency base and a high-frequency variability using a discrete wavelet transformation. Their method preserves the relationship between peak power and energy by sampling the two-dimensional distribution between them. They also apply their method to the problem of EV hosting capacity, which is very relevant to our case.

Deep learning techniques have also been considered for load modeling, in contrast to probabilistic models. [19] trains a neural network using a combination of measurements of consumption, weather data, and TOUS data to generate profiles of appliances and end-users. The authors of [18] use the same wavelet decomposition in their more recent work [20], but use a generative adversarial network (GAN) instead of a statistical model. Finally, an open-access tool that generates EV, PV, and load demand using GANs is described in [21]. Compared with stochastic models, deep learning models require extensive hyperparameter tuning and are harder to interpret.

The method proposed in this paper uses highly temporally resolved household measurements to train a statistical model capable of generating realistic load scenarios for a distribution network with given peak nodal demands, while preserving the statistical properties of the original dataset. The approach first clusters households based on their average demand profiles to identify distinct consumption patterns. The cluster means are then removed, and Gaussian models are fitted to the residuals to learn their time-dependent distributions. Different model configurations are evaluated based on their ability to reproduce key statistical properties of the original data, including mean and median demand profiles, as well as the peak-power-to-energy distribution, which is particularly relevant for BESS sizing applications. Finally, the selected model is validated on a BESS sizing problem and compared against an approach based on aggregated demand profiles.

The contributions of our method compared to the state of the art are the following. The proposed data-driven stochastic model trains a multi-dimensional GMM (as opposed to the one-dimensional used in [16, 22]) to model the joint distribution of the residuals of demand relative to the baseline. Compared with Markov-decision processes used in [15, 16], which model time steps individually, our method captures correlations between consumption at lags greater than 1. We propose two methods for modeling residuals, namely proportional and logarithmic residuals, and demonstrate, using available data, that logarithmic residuals can be accurately modeled by GMM. We then compute the distribution of the daily peak-power-to-energy for the synthetic scenarios and show that, by clipping extreme values generated by the model, we can reproduce the distribution of the original data. We finally validate the scenarios on a BESS sizing problem and show that the synthetic scenarios yield lower energy capacity and investment costs for the same power requirements than an aggregated model.

The rest of the paper is organized as follows: Section 2 presents the mathematical formulation of the scenario generation method; Section 3 validates the statistical properties of the generated scenarios; Section 4 presents an application of the generated scenarios on the optimal sizing of BESSs and validates the generated scenarios on said problem; Section 5 concludes the paper.

2 Mathematical Formulation

2.1 Notation and Problem Statement

We consider the time series of the power consumption of N households, labeled as $i \in \{1, 2, \ldots, N\}$ available for days $d \in \{1, 2, \ldots, D_i\}$ measured every ΔT minutes. We denote by $P_{i,d}(t)$ the power consumption in kW of household i in the interval $[t, t + \Delta T]$, where $t \in \{0, \Delta T, \ldots, 1440 - \Delta T\}$ denotes discretized time in minutes. We compute the per-unit (pu) power profiles rescaled according to the peak power of the household as follows:

$$\hat{P}_{i,d}(t) = \frac{P_{i,d}(t)}{\max_{d,t} P_{i,d}(t)}, \tag{1}$$

We denote by $\hat{\mathbf{P}}_{i,d} \in \mathbb{R}^{\frac{1440}{\Delta T}}$ the vector of all $\hat{P}_{i,d}(t)$ for $t \in \{0, \Delta T, \ldots, 1440 - \Delta T\}$. Our goal is to generate M synthetic load scenarios $\tilde{\mathbf{P}}_j \in \mathbb{R}^{\frac{1440}{\Delta T}}$, $j \in \{1, 2, \ldots, M\}$ with similar statistical properties as $\hat{\mathbf{P}}_{i,d}$.

2.2 Household Clustering

The first step in generating synthetic scenarios is to cluster households based on their average daily consumption profiles. This allows training a separate model for each cluster, which is expected to better model the statistical properties of households with different demand patterns, rather than training a single model for all of them. We consider the average daily demand of house i

$$\bar{\mathbf{P}}_i = \frac{1}{D_i} \sum_{d=1}^{D_i} \hat{\mathbf{P}}_{i,d}. \tag{2}$$

This leads us to a large-dimensional space with $1440/\Delta T$ dimensions. To reduce noise during the clustering, we reduce the dimensionality of the space using Principal Component Analysis (PCA) [23]. PCA is a widely used statistical method that transforms the original variables (timesteps in our case) into fewer variables (principal components) that best explain the variance in the original data. We first normalize the average demands $\bar{\mathbf{P}}_i$ so that they have zero mean and a standard deviation of 1 as required by PCA. We then run PCA to identify the L first principal components that are able to explain 95% of the variance of the data. This reduces the dimension of the data space from $1440/\Delta T$ to L.

We then perform K-means clustering in the reduced space of the normalized data to group the houses into K clusters based on their consumption patterns. We denote by $C_c = \{i : \mathcal{L}(i) = c\}$ the set of houses that belong to cluster c where $\mathcal{L}(i)$ is the cluster of house i. Then, we train a model for each of the identified clusters, as explained next.

2.3 Training of Statistical Model

A statistical model is trained independently for each identified cluster. To do so, the measurements belonging to a cluster are first detrended using the cluster's average demand profile. The underlying assumption is that the cluster average provides a sufficient predictor of the typical intra-day consumption trend, while the remaining residuals capture the stochastic variations around this trend.

The trend of each cluster c, $\hat{\mathbf{P}}^c_{trend}$, is given by averaging the demand profiles of all houses in cluster c for all days in $\mathcal{D}$:

$$\hat{\mathbf{P}}^c_{trend} = \frac{1}{|C_c| \cdot |\mathcal{D}|} \sum_{i \in C_c} \sum_{d \in \mathcal{D}} \hat{\mathbf{P}}_{i,d}, \tag{3}$$

where $|\cdot|$ denotes the cardinality of a set. Then, two methods are used for computing the residuals around the trend of a cluster.

Proportional residuals:

$$\hat{\mathbf{R}}^c_{i,d} = \frac{\hat{\mathbf{P}}^c_{i,d}}{\hat{\mathbf{P}}^c_{trend}}, \mathcal{L}(i) = c, d \in \mathcal{D}. \tag{4}$$

Logarithmic residuals:

$$\hat{\mathbf{R}}^c_{i,d} = \log\left(\frac{\hat{\mathbf{P}}^c_{i,d}}{\hat{\mathbf{P}}^c_{trend}}\right), \mathcal{L}(i) = c, d \in \mathcal{D}, \tag{5}$$

where log denotes the natural log.

The logarithmic residuals are always nonnegative, which provides an advantage for subsequent modeling steps compared to proportional residuals, which may take negative values. On the other hand, recovering the original signal requires exponentiation, which can amplify modeling errors and lead to extreme estimates. The two types of residuals are evaluated in Section 3 based on their ability to model the original data.

2.4 Generating Scenarios of Daily Demand in a Distribution Grid

We train a statistical model on the time-series residuals, using either a proportional or a logarithmic transformation. Various statistical models could be considered depending on the statistical properties of the original dataset. For this work, we consider Gaussian Mixture Models (GMMs) [24] of varying numbers of components because they accurately represent the available dataset, as will be seen in the validation section. Note that we train a model of size $1440/\Delta T$ for each cluster to capture temporal correlations in the time series.

Given M nodes in a distribution grid with nominal nodal powers $P^{\max}_j$, $j \in \{1, 2, \ldots, M\}$, we generate scenarios of daily demand for each node as follows:

(1) Assign each node j to a random cluster c with probability $|C_c|/N$.
(2) Sample a residual profile from the Gaussian mixture model (GMM) associated with cluster c, denoted by $\tilde{\mathbf{R}}_j$.
(3) Clip the sampled residuals to the interval $[\mathbf{R}^c_{LB}, \mathbf{R}^c_{UB}]$ where $\mathbf{R}^c_{LB}$ and $\mathbf{R}^c_{UB}$ are respectively the p_{LB}- and p_{UB}-percentiles of the original residuals $\hat{\mathbf{R}}^c_{i,d}$ belonging to cluster c. The purpose of this step is to prevent negative or extreme values in the generated demand profiles, sampled from the tails of the Gaussian distributions.
(4) Construct the demand profile for node j. For proportional residuals:

$$\tilde{\mathbf{P}}_j = P^{\max}_j \hat{\mathbf{P}}^c_{trend} \tilde{\mathbf{R}}_j, \tag{6}$$

and for logarithmic residuals:

$$\tilde{\mathbf{P}}_j = P^{\max}_j \hat{\mathbf{P}}^c_{trend} e^{\tilde{\mathbf{R}}_j}. \tag{7}$$

The logarithmic formulation guarantees strictly positive demand profiles.
(5) Repeat the residual sampling process to generate as many independent demand scenarios per node as required.

The process for generating scenarios for a given distribution grid is formally summarized in Algorithm 1

Algorithm 1 Process of scenario generation for a given distribution grid.

Input: (i) Dataset of pu of peak-power daily demand profiles $\hat{\mathbf{P}}_{i,d}$ for a given set of days $\mathcal{D}$, (ii) number of nodes M and maximum powers of nodes P^{max}_j of the distribution grid, (iii) choice of statistical model (number n of components of GMM, proportional or logarithmic residuals, percentiles of residuals to consider p_{LB}, p_{UB}).
Output: Nodal scenarios $\tilde{\mathbf{P}}_j$ scaled according to the maximum powers.
Cluster the houses in the dataset according to their average daily profile.
for all clusters c **do**
    Compute the trend of cluster (Eq. (3)).
    Compute the residuals of the houses in cluster c (Eq. (4) or (5)).
    Calculate the p_{LB}-th and p_{UB}-th percentiles of the residuals, $\mathbf{R}^c_{LB}$ and $\mathbf{R}^c_{UB}$ respectively.
    Train a GMM of order n on the residuals.
end for
for all nodes j **do**
    Sample a cluster c at random.
    Generate a scenario of the residuals $\tilde{\mathbf{R}}_j$ from the GMM of cluster c and clip in the range $[\mathbf{R}^c_{LB}, \mathbf{R}^c_{UB}]$.
    Compute the scaled scenario of the node (Eq. (6) or (7)).
end for
return Nodal scenarios $\tilde{\mathbf{P}}_j$.

3 Training and Validation

3.1 Training Data

We train our model on the HEAPO dataset [14], an open dataset of household consumption originally assembled for heat pump optimization. It contains measurements of base load, heat pump consumption, and photovoltaic (PV) generation from 1408 households in the Canton of Zurich, Switzerland. Measurements are available at 15-minute resolution from 2018-11-03 to 2024-03-21, with each house having, on average, 638.75 days of available data. It also contains useful metadata on household characteristics reported by the residents, such as the presence of electric vehicles (EVs), dryers, freezers, floor heating, etc., and more detailed data collected during on-site visits on certain households, such as the heating capacity of the heat pump and the size of the installed PV system.

The dataset is particularly suitable for our study, as it captures a heterogeneous mix of demand and distributed energy resources. Since the proposed method is primarily designed to model stochastic loads, heat pump consumption and PV generation are excluded, and only base-load measurements are considered. The reason for this choice is that the proposed approach is essentially a black-box model intended for consumption patterns that are difficult to describe using physical models. In contrast, heat pump consumption is associated with well-understood physical processes and can be accurately modeled using a physical model, as in [25, 26]. A similar consideration applies to PV generation, which can generally be represented accurately using physics-based models [27].

We rescale measurements for each household to pu of the maximum power consumption, according to Eq. (1), to obtain the rescaled daily profiles $\hat{P}_{i,d}(t)$. Fig. 1 shows the average pu daily demand profile of K=5 clusters obtained from K-means clustering on the 1408 households of the HEAPO dataset using all the available days, as described in Section 2.2. It also shows the percentage of households belonging to each cluster. Clusters 3 and 5, which together account for nearly half of the households, exhibit a typical residential demand profile characterized by an evening peak, similar to aggregated profiles such as the load profile proposed by CIGRE [28], commonly adopted in power system studies. Cluster 4 also exhibits an evening peak around 6 pm, together with a secondary peak around 6 am. Finally, Cluster 2 shows a sharp peak around midnight, while Cluster 1 exhibits a pronounced peak around 2 am, which may correspond to loads scheduled to operate at night, such as water heaters or residual electric heating.

To train and evaluate the statistical model, we split the households into training and test datasets by randomly sampling 80% of the households for training and 20% for testing. Our model is trained on the training dataset using Algorithm 1. Then, 10,000 scenarios of the pu demands are generated, and their statistical properties are compared with those of the test dataset using the performance metrics presented in the next section.

3.2 Performance Metrics

We consider the vectors $\hat{\mathbf{P}}_{i,d} \in \mathbb{R}^{\frac{1440}{\Delta T}}$ of the original (test) daily profiles for days $d \in \mathcal{D}$ and buildings $i \in \{1, 2, \ldots, N_{\text{test}}\}$. We also consider the vectors $\tilde{\mathbf{P}}_j \in \mathbb{R}^{\frac{1440}{\Delta T}}$ of the M generated synthetic daily

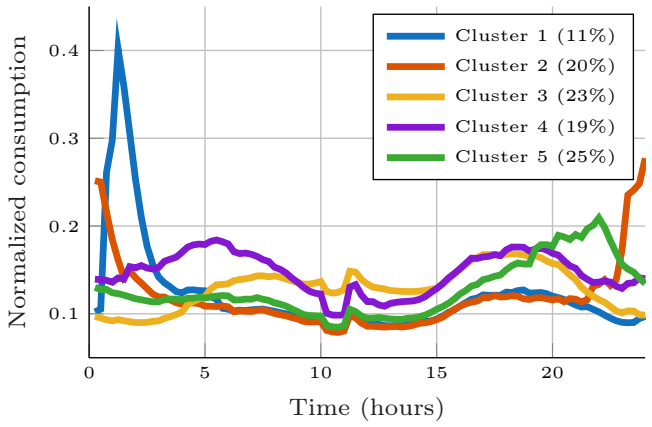


Fig. 1. Average pu demand profile per cluster obtained from K-means clustering, showing the percentage of houses in each cluster.

profiles, $j \in \{1, 2, \ldots, M\}$. The following metrics quantify the difference between statistical properties of the original measurements $\hat{\mathbf{P}}_{i,d}$ and the generated scenarios $\tilde{\mathbf{P}}_j$.

(1) *Mean and median daily profile*: The first two metrics quantify the percentage difference between the average and median profiles of the original measurements and those of the generated scenarios. The metric for the mean is:

$$m_{\text{mean}} = \frac{\|\hat{\mu} - \tilde{\mu}\|_2}{\|\hat{\mu}\|_2} \times 100\%, \tag{8}$$

where

$$\hat{\mu} = \frac{1}{N_{\text{test}} \cdot |\mathcal{D}|} \sum_{i=1}^{N_{\text{test}}} \sum_{d \in \mathcal{D}} \hat{\mathbf{P}}_{i,d} \tag{9}$$

is the average consumption profile of the test dataset over all houses and days, and

$$\tilde{\mu} = \frac{1}{M} \sum_{j=1}^{M} \tilde{\mathbf{P}}_j \tag{10}$$

is the average consumption profile of the generated scenarios. Similarly, the metric for the median is:

$$m_{\text{med}} = \frac{\left\|\text{med}_{i,d}(\hat{\mathbf{P}}_{i,d}) - \text{med}_j(\tilde{\mathbf{P}}_j)\right\|_2}{\left\|\text{med}_{i,d}(\hat{\mathbf{P}}_{i,d})\right\|_2} \times 100\%, \tag{11}$$

where $\text{med}_{i,d}(\hat{\mathbf{P}}_{i,d})$ denotes the median profile of $\hat{\mathbf{P}}_{i,d}$ across all houses i and days d and $\text{med}_j(\tilde{\mathbf{P}}_j)$ denotes the median profile across all scenarios j. Note that the norms in Eq. (8) and (11) are both applied in a vector in $\mathbb{R}^{\frac{1440}{\Delta T}}$.

(2) *Correlation matrix*: It measures how well the model captures the inter-temporal correlation of the original data:

$$m_{\text{corr}} = \frac{\left\|\text{Corr}(\hat{\mathbf{P}}_{i,d}) - \text{Corr}(\tilde{\mathbf{P}}_j)\right\|}{\left\|\text{Corr}(\hat{\mathbf{P}}_{i,d})\right\|} \times 100\%, \tag{12}$$

where $\text{Corr}(\cdot)$ denotes the correlation matrix, of dimensions $\frac{1440}{\Delta T} \times \frac{1440}{\Delta T}$, and the norm indicates the Frobenius norm.

(3) *Distribution of the power-to-energy ratio*: It measures how well the model captures peak powers during the day. It is quantified by the p-th order Wasserstein distance [29] between the Empirical Cumulative Distribution Functions (ECDFs) of the daily peak-power-to-energy ratio of the rescaled original data $\hat{F}(x)$ and of the generated scenarios $\tilde{F}(x)$, given as a percentage of the p-th order norm of $\hat{F}^{-1}(x)$:

$$W_p = \frac{\left(\int_0^1 \left|\hat{F}^{-1}(x) - \tilde{F}^{-1}(x)\right|^p \, \mathrm{d}x\right)^{\frac{1}{p}}}{\left(\int_0^1 \left|\hat{F}^{-1}(x)\right|^p \, \mathrm{d}x\right)^{\frac{1}{p}}} \times 100\%. \tag{13}$$

Higher orders p penalize large deviations between the two distributions.

In the following sections, we compute the above metrics by varying the parameters of the statistical model to determine the model that best approximates the original data. The metrics are computed on the test data to assess the model's ability to generalize to data not used for training.

3.3 Comparison Between Proportional and Logarithmic Residuals

We first compare the proportional and logarithmic residuals in terms of their ability to approximate the mean and median daily demand profiles. For this section, we group the houses into five clusters, as shown in Fig. 1 (the effect of the number of clusters will be studied in Section 3.7). Fig. 2 shows the proportional and logarithmic residuals of the training dataset at a specific time index corresponding to 12 pm. We infer that the logarithmic residuals for a given timestep can be accurately represented by a GMM of order 2, given the presence of 2 peaks in Fig. 2b. On the other hand, the proportional residuals are not accurately represented by a Gaussian because the residuals of Fig. 2a are not symmetric and non-negative.

To verify this, we compare the mean and the median daily profile of the test data and the generated scenarios. Figs. 3a and 3b show the mean and median profiles of scenarios generated using proportional and logarithmic residuals, respectively, and a GMM of order 2 compared to the test dataset. The proportional residuals can accurately model the mean demand but fail to capture the median. Indeed, the median of the original data is smaller than the mean, as is also verified by the histogram of Fig. 2a, hence they cannot be modeled by a Gaussian. Moreover, although the GMM can model the mean demand, it may yield negative demand in certain scenarios, which must be filtered out. On the other hand, training a GMM on the logarithmic residuals accurately captures the median profile while ensuring strictly positive profiles. However, the model results in a significantly higher mean profile than the original data. This is due to the exponent in Eq. (7), which can yield extreme values that inflate the mean but do not affect the median.

To prevent negative values on the proportional residuals and extreme values on the logarithmic residuals, we clip all generated residuals between the p_{LB} = 0th and p_{UB} = 99th percentile of the respective residuals of the original data. Figs. 3c and 3d show the mean and median profiles obtained by clipping the proportional and logarithmic residuals, respectively. Removing the negative values

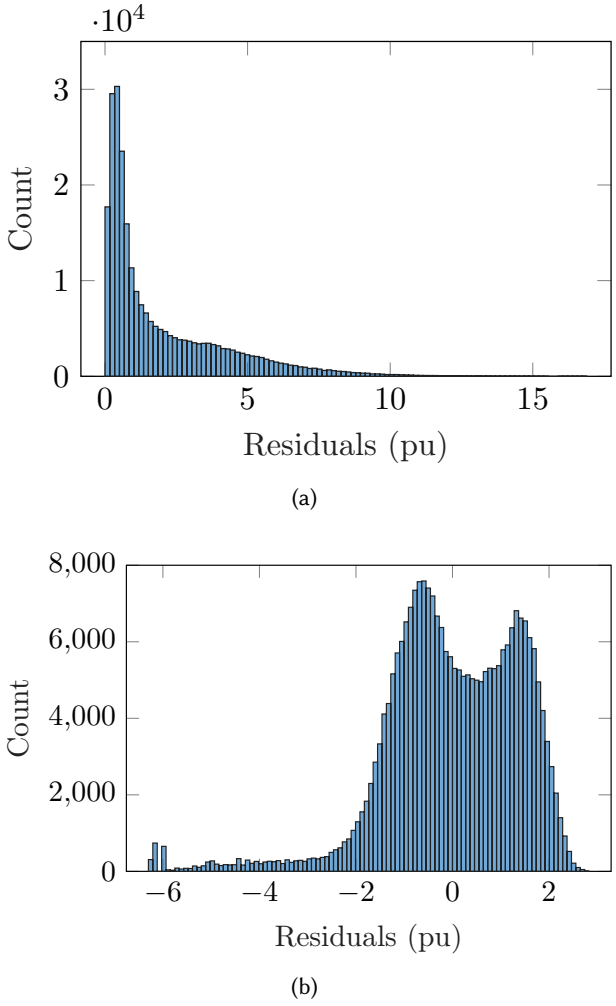


Fig. 2. Histogram of (a) proportional and (b) logarithmic residuals of the training dataset at 12 pm.

obtained from the proportional residuals inflates the mean profile, so the model no longer captures either the mean or the median. On the other hand, clipping the extreme values of the logarithmic residuals shrinks the mean profile, which now matches that of the test dataset, without affecting the median. Therefore, we conclude that clipping logarithmic residuals is the most effective model for capturing both the mean and median profiles and generalizes well, accurately representing data it has not seen.

3.4 Peak-Power-to-Energy Ratio

It has been shown that training a model on the logarithm of the residuals can effectively capture the mean and median of the original dataset. However, modeling the average demand profile is insufficient because it does not capture extreme daytime peaks, which might generate grid congestion or violate grid voltage constraints.

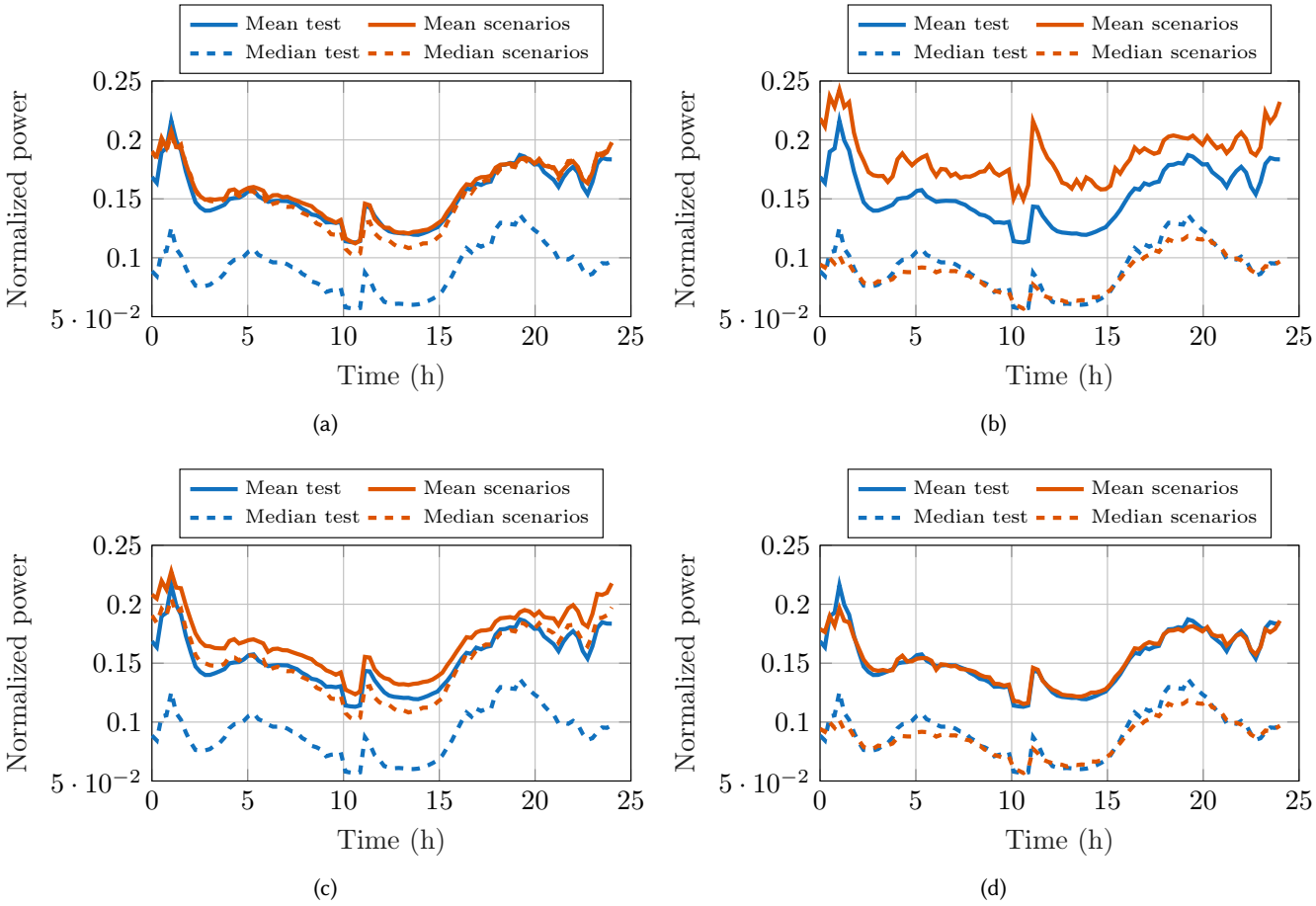


Fig. 3. Mean and median daily profiles of test dataset and generated scenarios for (a) proportional and (b) logarithmic residuals without clipping extreme values, and (c) proportional and (d) logarithmic residuals with clipping extreme values between the 0th and 99th percentile.

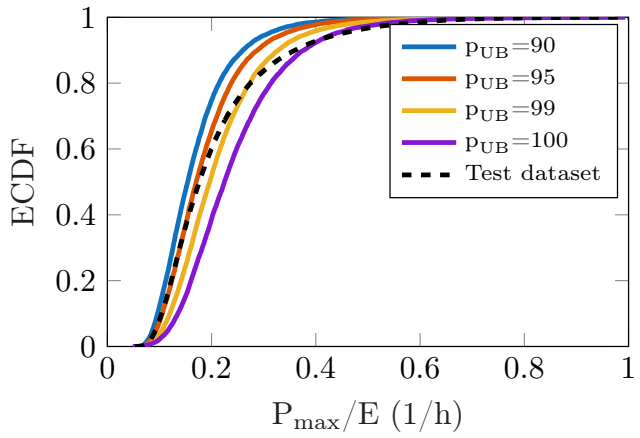
To assess whether the model accurately reproduces peaks, we compute the ECDF of the daily peak-power-to-energy ratio for the test dataset and the generated scenario. Fig. 4a shows the ECDF for different values of the upper-bound percentile p_{UB} used to clip the logarithmic residuals for a GMM order of 2. If we clip the peaks too low, the model yields smaller peaks at the same energy compared to the original data. On the other hand, if we clip at the maximum observed residual, the model captures extreme peaks well but consistently yields higher peaks at the same energy. Choosing an upper bound between the 95th and 99th percentiles yields an ECDF closer to that of the original data.

To quantify the difference between the ECDf of the original and synthetic profiles, we compute the 1st- and 2nd-order percentage Wasserstein distances between the two. Fig. 4b plots W_1 and W_2 as a function of the chosen upper bound of the logarithmic residuals p_{UB}. The percentage distance between the two ECDFs is minimal at $p_{\text{UB}} = 97$, as this value minimizes W_1. However, $p_{\text{UB}} = 99$ more closely matches the ECDF if large differences are penalized, as it is the value that minimizes W_2. In what follows, we continue to use $p_{\text{UB}} = 99$ to penalize large modeling errors in the peak powers.

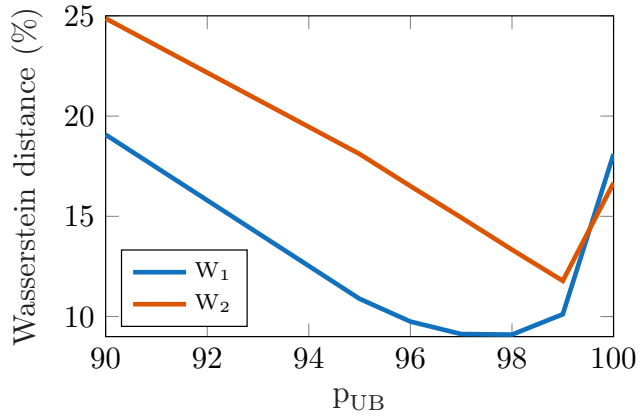
3.5 Metrics Evaluation

We next evaluate the metrics presented in Section 3.2 for the dominant-so-far model, which trains a GMM of order 2 on 5 clusters using the logarithmic residuals clipped to the 99th percentile of the training data. The metrics will serve as benchmarks for the following sections to assess whether higher GMM orders or more clusters improve the algorithm's performance. We also compare against a simple baseline that could be considered for a robust optimization problem consisting of 4 scenarios, namely (i) the mean profile of the training dataset, (ii) the median profile, (iii) the 5th percentile, and (iv) the 95th percentile.

Table 1 shows the values of the various metrics for this simple baseline and the dominant model. We observe reductions of almost 8-fold in the mean metric and 5-fold in the median metric compared to the baseline. The improvements are due to the baseline's assumption



(a)



(b)

Fig. 4. (a) ECDF of daily peak-power-to-energy ratio for different values of the upper-bound percentile p_{UB} of the logarithmic residuals. (b) 1st- and 2nd-order Wasserstein distances between the test dataset and the generated scenarios.

Model	m_{mean}%	m_{med} %	m_{corr} %	W_2 %
Baseline	21.9	37.0	326	78.5
Dominant model	2.81	7.89	23.6	11.8

Table 1. Comparison of metrics between a simple baseline and the dominant model of GMM of order 2 using log residuals clipped to the 99th percentile.

that the load profiles are symmetric around the mean, which is invalid, as verified in Section 3.3.

The biggest improvement from the baseline is in the correlation metric, which is 14 times smaller using our model. This is because the baseline uses average scenarios, which smooth out temporal variations in household demand, resulting in a much higher correlation between timesteps. Even though our model significantly improves the modeling of inter-temporal correlation, the metric of

GMM order	m_{mean}%	m_{med} %	m_{corr} %	W_2 %
1	3.03	8.02	23.3	12.6
2	2.81	7.89	23.6	11.8
3	2.59	7.46	23.6	11.4
4	2.77	7.18	24.3	10.9
5	2.83	8.19	23.3	11.5

Table 2. Evaluation metrics for log residuals with 99th percentile for different GMM orders.

correlation matrix inaccuracy remains high, at 23% of the original dataset's norm. This means that the GMM cannot model the correlation matrix as well as the mean and median demand profiles. More sophisticated models could be studied in the future to better capture the intertemporal characteristics of household demand.

Finally, our model improves the relative Wasserstein distance W_2 for the peak-power-to-energy ratio by a factor of 7. This is also expected, as the average scenarios used in the baseline reduce peak power, resulting in a distribution that deviates significantly from the original data.

3.6 Effect of GMM Order

In this section, we quantify the benefit of training a model with a higher-order GMM. We use 5 clusters for the households and clip the logarithmic residuals to the 99th percentile. Table 2 compares the various metrics for varying GMM orders. A GMM order of 3 outperforms order 1 across the mean, median, and Wasserstein distance metrics, with improvements ranging from 6-14% per metric, whereas the performance for the correlation metric is marginally worse. Order 4 is better than 3 on the median and Wasserstein metrics, but performs worse on the mean and covariance metrics. However, all metric differences are marginal, so we can safely conclude that the GMM order is not a critical factor in the model's performance. This happens because we model a high-dimensional variable (96 dimensions), so we might need a very complex model. If the peaks in the residuals of Fig. 2b differ slightly for each timestep, then we cannot fit a multivariate Gaussian variable with confidence. On the other hand, higher-order models are harder to train and are more likely to overfit the training data, leading to worse performance on the test set.

3.7 Effect of Number of Clusters

We finally study the effect of the number of household clusters on the model's performance. We train the same model as in the previous section, using a GMM order of 4. Table 3 shows the various metrics for different numbers of clusters. We observe that varying the number of clusters can improve some metrics at the expense of others, and that there is no measurable benefit to using one number of clusters over another. This contradicts our initial hypothesis that training a separate model for each cluster would improve the accuracy of the statistical properties of the scenarios. Nevertheless, generating scenarios from different clusters is a useful feature of the algorithm because it can produce scenarios with peaks at specific times, which more accurately represent the demand patterns of individual households.

# of clusters	m_{mean} %	m_{med} %	m_{corr} %	W_2 %
1	2.59	7.34	22.8	12.6
2	2.54	7.44	23.4	11.6
3	2.50	7.18	24.3	11.6
4	2.67	7.26	23.5	12.1
5	2.77	7.18	24.3	10.9
6	2.88	7.72	22.3	11.7
7	2.92	8.18	21.7	10.7
8	2.94	7.80	21.8	10.9
9	2.63	7.76	22.1	10.4
10	2.66	7.41	22.5	11.3

Table 3. Evaluation metrics for log residuals with 99th percentile with GMM order of 4 for different numbers of clusters.

4 Validation on the Sizing of Batteries

As an application of the proposed model, this section uses it to estimate the optimal siting and sizing of batteries in a low-voltage distribution grid as an alternative to grid reinforcement. The approach is based on a standard optimal power flow formulation, which is explained in detail next for clarity.

The goal is to determine the optimal locations (siting) and energy capacities (sizing) of BESSs in a given network to minimize total investment costs while respecting the grid's constraints. We compute the investment costs associated with the nodal scenarios generated by our method and compare them with the costs under the CIGRE profile for all nodes. Sections 4.1-4.5 formulate the sizing optimization problem, Section 4.6 presents the case study for the sizing problem, and Section 4.7 presents the results obtained using our scenarios compared to the CIGRE profile. Section 4.8 discusses the results and potential future work.

4.1 Cost Function

We denote by $\mathcal{N}$ the set of nodes in a distribution grid. E_n is the energy capacity of the BESS to be installed at node $n \in \mathcal{N}$, and S_n is the power rating of the converter interfacing the BESS to the grid. We denote by $I_n \in \{0, 1\}$ the installation variable of node n, where $I_n = 1$ if a BESS is installed at node n and 0 otherwise.

The total investment cost comprises the energy capacity, converter, and installation costs. The energy capacity cost is defined as

$$\mathcal{F}^{\mathrm{E}} = \sum_{n \in \mathcal{N}} c^{\mathrm{E}} E_n, \tag{14a}$$

where c^{E} is the price of energy capacity in \$/kWh. Similarly, the converter cost is defined as

$$\mathcal{F}^{\mathrm{S}} = \sum_{n \in \mathcal{N}} c^{\mathrm{S}} S_n, \tag{14b}$$

where c^{S} is the price of the converter in \$/kVA. Finally, the installation cost is defined as

$$\mathcal{F}^{\mathrm{I}} = \sum_{n \in \mathcal{N}} c^{\mathrm{I}} I_n, \tag{14c}$$

where c^{I} is the cost in \$ of installing a BESS.

4.2 BESS Constraints

We consider the time index $t \in \{0, \Delta T, \ldots, T\}$, where ΔT is the timestep of available measurements (15 minutes in the HEAPO dataset), and T is the time horizon of 24 hours. We denote by $P_n^{\mathrm{B}+}(t)$, $P_n^{\mathrm{B}-}(t)$ the charging and discharging powers, respectively, and by $P_n^{\mathrm{B}}(t)$ the power injection of the BESS at node n, with positive power indicating charging. The charging and discharging power should satisfy the following constraints:

$$P_n^{\mathrm{B}}(t) = P_n^{\mathrm{B}+}(t) - P_n^{\mathrm{B}-}(t) \tag{15a}$$

$$P_n^{\mathrm{B}+}(t), P_n^{\mathrm{B}-}(t) \geq 0 \tag{15b}$$

$$P_n^{\mathrm{B}+}(t) \cdot P_n^{\mathrm{B}-}(t) = 0 \tag{15c}$$

Eq. (15c) models the exclusivity of the two powers that prevents simultaneous charging and discharging of the BESS. The State of Energy (SOE) of the BESS is characterized by the following model

$$\mathrm{SOE}_n(t) = \mathrm{SOE}_n(t-1) + \Delta T \left(\eta^{\mathrm{ch}} P_n^{\mathrm{B}+}(t) - \frac{1}{\eta^{\mathrm{dis}}} P_n^{\mathrm{B}-}(t) \right), \tag{15d}$$

where $\eta^{\mathrm{ch}}, \eta^{\mathrm{dis}}$ are the charging and discharging efficiencies of the BESS, respectively. We require that the SOE remain below the respective energy capacity. We also enforce that the SOE at the end of the day equals that at the beginning of the day to ensure sufficient energy to serve the following day. These are enforced by the following constraints.

$$0 \leq \mathrm{SOE}_n(t) \leq E_n, \tag{15e}$$

$$\mathrm{SOE}_n(0) = \mathrm{SOE}_n(24\mathrm{h}), \tag{15f}$$

The maximum power of the BESS is linked to its energy capacity by the C-rate, denoted by C^{rate}, which is technology-dependent, as follows:

$$P_n^{\mathrm{B}+}(t), P_n^{\mathrm{B}-}(t) \leq C^{\mathrm{rate}} E_n \tag{15g}$$

Finally, the BESS power is limited by the power converter's capability curve, which is approximated by a box for simplicity, as follows:

$$P_n^{\mathrm{B}+}(t), P_n^{\mathrm{B}-}(t) \leq k_P S_n \tag{15h}$$

$$|Q_n^{\mathrm{B}}(t)| \leq k_Q S_n, \tag{15i}$$

where $k_P, k_Q \geq 0$ are design parameters chosen such that $k_P^2 + k_Q^2 \leq 1$ that decide how much reactive power should be limited. In our case, we choose $k_P = 0.98$ to restrict reactive power production.

It should be noted that (15c) is non-convex. To ensure convexity, we approximate the mutual exclusivity of charging and discharging by adding a penalty term to the cost function for $P_n^{\mathrm{B}+}(t)$ and $P_n^{\mathrm{B}-}(t)$ to penalize simultaneous charging and discharging.

4.3 Installation constraints

To model the installation of BESSs, we define the following constraints:

$$E_n \leq E^{\max} I_n \tag{16a}$$

$$S_n \leq S^{\max} I_n, \tag{16b}$$

where $E^{\max}, S^{\max}$ are the maximum allowed energy capacity and converter rating, which can be chosen as very high numbers.

4.4 Grid Constraints

The net nodal power injections are as follows:

$$P_n(t) = P_n^{\mathrm{B}}(t) + P_n^{\mathrm{L}}(t) - P_n^{\mathrm{PV}}(t) \tag{17a}$$

$$Q_n(t) = Q_n^{\mathrm{B}}(t) + Q_n^{\mathrm{L}}(t) - Q_n^{\mathrm{PV}}(t), \tag{17b}$$

where $P_n^{\mathrm{L}}(t)$, $Q_n^{\mathrm{L}}(t)$ are the active and reactive loads, and $P_n^{\mathrm{PV}}(t)$, $Q_n^{\mathrm{PV}}(t)$ are the active and reactive power of the PV connected to node n. The load and PV generation are fixed, whereas the BESS power is computed by the optimization algorithm. The active power of the load at each node is computed by sampling a random scenario with our method and scaling it by the node's maximum active power. The load reactive power is computed from the active power using a fixed power factor of 0.9. For the PV active power, we use the same sunny-day profile for all nodes, and we assume that all reactive power is 0. We then impose further grid constraints on the net active and reactive power injections using a linearized grid model, as shown below.

4.4.1 Linear grid model. We use the method of sensitivity coefficients [30] to construct a linear grid model, yielding a convex formulation of the sizing problem. We consider the initial nodal power injections before installing the BESSs $P_n^0(t) = P_n^L(t) - P_n^{PV}(t)$ and $Q_n^0(t) = Q_n^L(t) - Q_n^{PV}(t)$. The sensitivity coefficients are the derivatives of the grid state (nodal voltages, branch currents, or slack power) with respect to net power injections. We define the power differentials $\Delta P_n(t) = P_n(t) - P_n^0(t)$ and $\Delta Q_n(t) = Q_n(t) - Q_n^0(t)$[1]. Then, the linearized grid model consists of the following linear equations:

(1) Nodal voltages $v_n(t)$:

$$v_n(t) = v_n^0(t) + \sum_{m\in\mathcal{N}} K_{v,nm}^P(t)\Delta P_m(t) + K_{v,nm}^Q(t)\Delta Q_m(t), \tag{17c}$$

where $v_n^0(t)$ is the voltage of node n computed at the points $P_n^0(t)$, $Q_n^0(t)$, and $K_{v,nm}^P(t)$, $K_{v,nm}^Q(t)$ are the sensitivity coefficients of the voltage at node n with respect to the active and reactive power, respectively, at node m. The sensitivity coefficients can be computed analytically according to the method in [30].

(2) Branch currents $i_l(t)$:

$$i_l(t) = i_l^0(t) + \sum_{m\in\mathcal{N}} K_{i,lm}^P(t)\Delta P_m(t) + K_{i,lm}^Q(t)\Delta Q_m(t), \tag{17d}$$

where $i_l^0(t)$ is the current at branch l computed at $P_n^0(t)$, $Q_n^0(t)$, and $K_{i,lm}^P(t)$, $K_{i,lm}^Q(t)$, are the current sensitivity coefficients.

(3) Slack active and reactive power $P_s(t)$, $Q_s(t)$:

$$P_s(t) = P_s^0(t) + \sum_{m\in\mathcal{N}} K_{P_s,m}^P(t)\Delta P_m(t) + K_{P_s,m}^Q(t)\Delta Q_m(t), \tag{17e}$$

$$Q_s(t) = Q_s^0(t) + \sum_{m\in\mathcal{N}} K_{Q_s,m}^P(t)\Delta P_m(t) + K_{Q_s,m}^Q(t)\Delta Q_m(t), \tag{17f}$$

where $P_s^0(t)$, $Q_s^0(t)$ are the active and reactive power at the slack bus computed at $P_n^0(t)$, $Q_n^0(t)$. $K_{P_s,m}^P(t)$, $K_{P_s,m}^Q(t)$ are the sensitivity coefficients of the slack's active power and $K_{Q_s,m}^P(t)$, $K_{Q_s,m}^Q(t)$ are the sensitivities of the slack's reactive power.

[1]Note that for our application $\Delta P_n(t) = P_n^B(t)$ and $\Delta Q_n(t) = Q_n^B(t)$.

Node	Fuse rating (kW)
28, 37-39	24.94
29	249.4
30	24.32
31	62.35
33-35	21.82

Table 4. Fuse ratings of nodes.

Given the linearized grid model, we can define the following grid constraints.

4.4.2 Voltage and current constraints. The voltages should remain between defined limits $[\underline{v}, \overline{v}]$ (e.g., [0.9, 1.1]pu). The branch currents should stay below the ampacity rating of the branch $\overline{i}_l$.

$$\underline{v} \le v_n(t) \le \overline{v}, \tag{17g}$$

$$0 \le i_l(t) \le \overline{i}_l. \tag{17h}$$

4.4.3 Slack bus constraints. The constraints for the slack bus are similar to the power convert constraints of a BESS (15h):

$$|P_s(t)| \le k_{P_s}\overline{S} \tag{17i}$$

$$|Q_s(t)| \le k_{Q_s}\overline{S}, \tag{17j}$$

where $\overline{S}$ is the apparent power rating of the transformer at the grid connection point and $k_{P_s}^2 + k_{Q_s}^2 \le 1$. Because our goal is to limit reactive power import/export with the utility grid, we set $K_P = 0.98$, as in Eq. (15h).

4.5 Optimization Problem Formulation

The full sizing and siting optimization problem is the following:

$$\min_{E_n, I_n} \mathcal{F}^{\mathrm{E}} + \mathcal{F}^{\mathrm{S}} + \mathcal{F}^{\mathrm{I}} + \alpha \sum_{n\in\mathcal{N}} \sum_{t=0}^{T} P_n^{\mathrm{B+}}(t) + P_n^{\mathrm{B-}}(t) \tag{18a}$$

$$\text{s.t. BESSs constraints (15)} \tag{18b}$$

$$\text{Installation constraints (16)} \tag{18c}$$

$$\text{Grid constraints (17)} \tag{18d}$$

We have added a linear penalty term with a small weight α on the charging and discharging of the BESSs to prevent simultaneous charging and discharging, as explained in Section 4.2. Because of the existence of the binary installation variables I_n and because all constraints are linear, Problem (18) is a mixed-integer linear problem (MILP), which can be solved by solvers such as Gurobi.

4.6 Case study

We consider an LV distribution grid located in the Canton of Valais, Switzerland, as shown in Fig. 5. It has 39 nodes, out of which 10 have demand. The maximum power of each node is determined by its fuse rating, which is the maximum power that can flow through the line connecting the end node to the grid. The fuse rating of each node with demand is shown in Table 4. The line connecting nodes

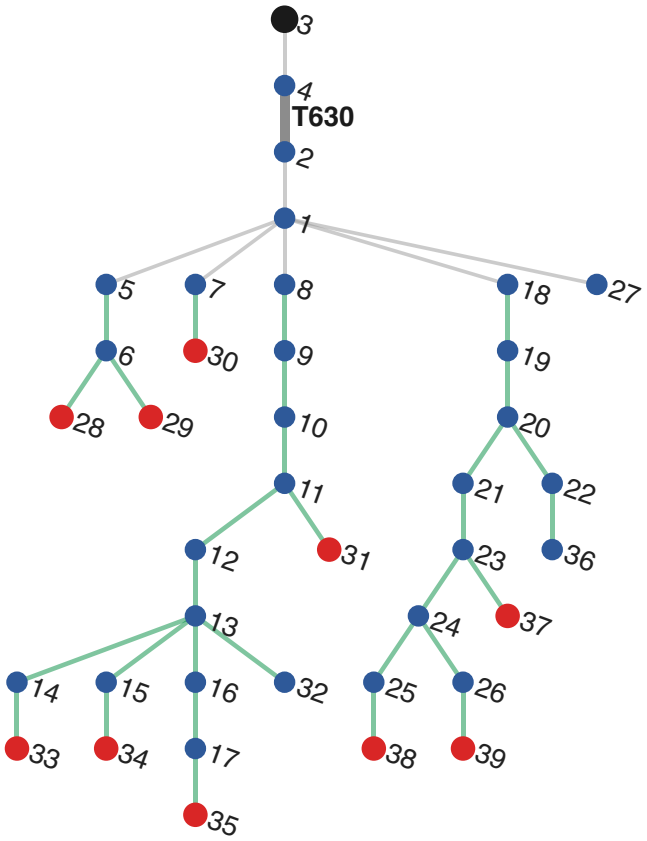


Fig. 5. Network topology for BESS sizing. The red nodes are the ones with demand present. Line 2-4 represents a 630 kVA MV/LV transformer, whereas the gray lines are artificial lines that model switches.

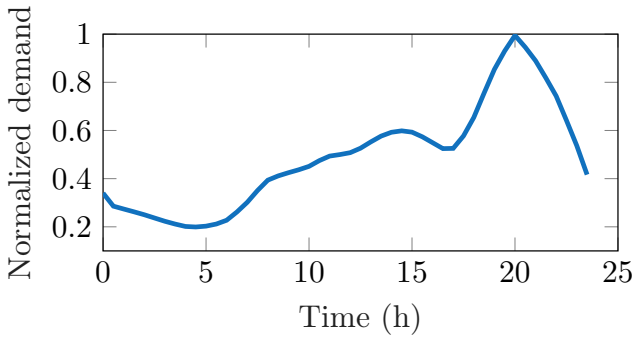


Fig. 6. CIGRE pu residential load profile.

2 and 4 represents a 630 kVA MV/LV transformer that connects the LV grid to the main grid.

We solve the sizing and siting optimization problem on the network using load scenarios generated by our method and compare the results with the CIGRE profile, as shown in Fig. 6. For a fair comparison, we generate scenarios using our method exclusively

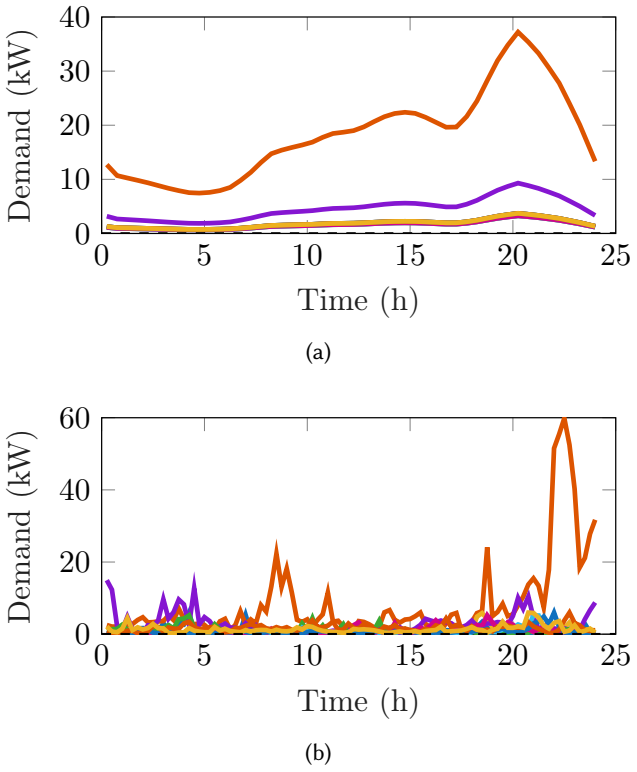


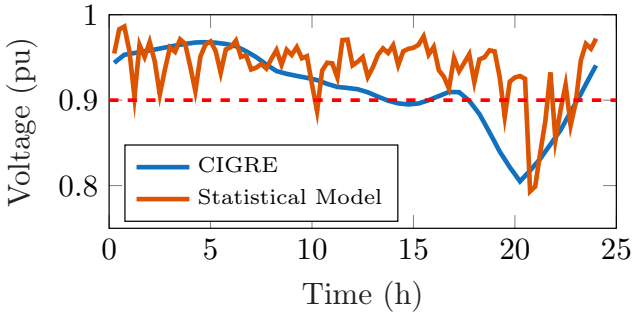
Fig. 7. Nodal demand profiles using (a) the CIGRE profile and (b) using our statistical model.

from Cluster 5 in Fig. 1, which has characteristics similar to the CIGRE profile, and both exhibit a peak demand in the evening (8-9 pm). For our method, we train a GMM of order 4 on the logarithmic residuals on summer weekdays and clip the residuals at the 99th percentile of the measurements.

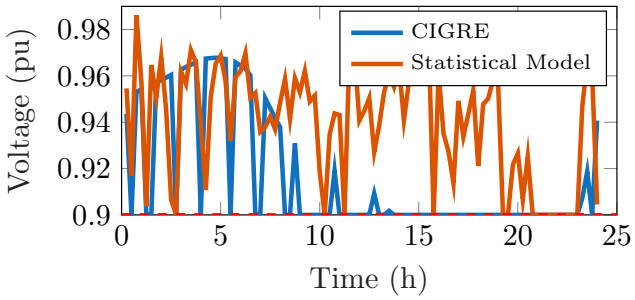
The scenarios of each node are scaled according to the node's fuse rating. In both the CIGRE profile and our scenarios, we scale the nodal demands as a percentage of the respective fuse ratings until the minimum voltage of 0.8 pu is reached. The load demands from both methods are shown in Fig. 7, and Fig. 8a shows the minimum nodal voltage before BESS installation for the two methods. Using the CIGRE profile for all nodes synchronizes the consumers' behavior, and the voltages reach 0.8 pu at 8 pm, with a maximum load of 15% of each node's respective fuse rating. On the other hand, the statistical model produces peaks at slightly different times, so a higher consumption, equal to 24% of the fuse rating, is needed to reach the same voltage level as the CIGRE profile. Moreover, the CIGRE profile exhibits higher evening energy consumption, necessitating greater energy capacity to regulate voltages, as will be verified in the following section.

4.7 Results

We run the BESS sizing optimization problem using the nodal demands of Fig. 7. We consider installing lithium-ion BESSs with a

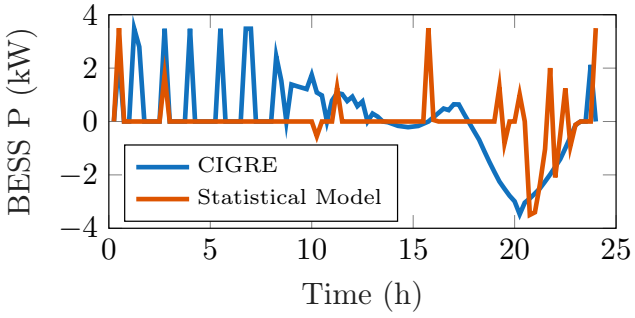


(a)

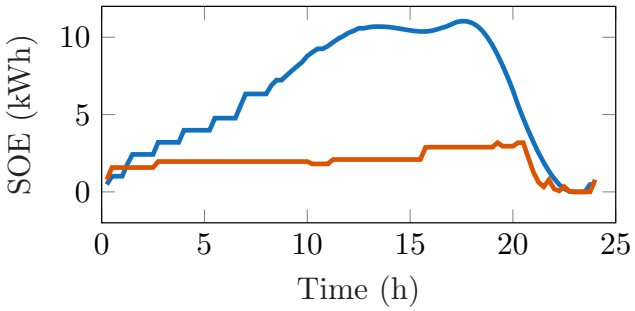


(b)

Fig. 8. Nodal voltages (a) before installing a BESS and (b) after optimal BESS sizing.



(a)



(b)

Fig. 9. (a) BESS active power and (b) BESS SOE.

Profiles	# BESSs	Node n	E_n (kWh)	S_n (kVA)	Cost ($)
CIGRE	1	39	11.04	3.55	5668
GMM	1	39	3.18	3.57	3311

Table 5. Results of sizing problem using the CIGRE profile and scenarios generated by GMM.

maximum C-rate of 1.1 and charging and discharging efficiencies of 0.9. We consider an energy capacity cost of 300 $/kWh, a converter cost of 100 $/kVA, and an installation cost of 2000 $. We formulate the MILP using CVX in MATLAB and solve it with the Gurobi solver. Fig. 8b shows the minimum voltage in the grid after installing the BESS. We observe that the BESS computed by the sizing algorithm meets the grid voltage limits in both cases.

Table. 5 shows the results of the sizing algorithm in the two cases of load profiles. In both cases, the algorithm computed a single BESS located at node 39, which has a demand far from the feeder. The converter's power is nearly identical in both cases because the nodal profiles were scaled to keep the total peak load demand the same. However, the statistical model requires 3.5 times less energy capacity for the same converter, thereby requiring 40% less investment to meet the same voltage regulations.

This is because the statistical model more accurately captures intra-day dynamics than the CIGRE profile, which represents an average demand profile. If we compare the CIGRE profile with the average pu demand of houses in Cluster 5 shown in Fig. 1, we observe that the latter has a peak demand that is 5 times smaller at 0.2 pu. This occurs because averaging smooths out the peaks without affecting daily energy consumption. The CIGRE profile represents an average demand profile, so normalizing to the 0-1 range inflates its energy demand; hence, the network requires greater energy capacity to maintain voltage balance. On the other hand, our statistical model accurately captures the daily peak-power-to-energy ratio of household demand, as demonstrated in Section 3.4, enabling the optimization to couple the network's energy and power demands more accurately.

The coupling between energy and power becomes clear by looking at the BESS power and SOE in the two cases shown in Fig. 9. In both cases, the BESS discharges at the same power in the evening to increase the voltages to the required bound; therefore, the required power converter is the same. However, for the CIGRE profile, the BESS requires a much higher energy reserve due to the inflated energy demand to anticipate the evening voltage drop.

4.8 Discussion and Future Work

The sizing problem results confirm that demand profiles obtained from a statistical model can reduce investment costs compared to an average profile, such as the CIGRE profile. This is expected because average profiles exhibit higher intertemporal correlation than real measurements and do not model peak powers, so they require more energy to balance the grid. For a more thorough evaluation of our

method, we should compare the sizing results against real demand profiles of the network's consumers, which were not available for the network considered in this case study. Another interesting study would be to evaluate the robustness of the sizing under different seeds of the scenario-generation algorithm. This would give us insight into whether a single scenario per node generated by our method is sufficient to compute a reliable estimate of the network's BESS needs, or whether multiple scenarios per node should be considered.

A limitation of the proposed statistical model is that it does not account for the spatial correlation of demand among nodes. Although the correlation of sampled demand profiles within a given node is implicitly captured by the clustering, our method samples one cluster independently for each node in the network. Spatial correlation in the scenario generation will be considered in the future.

As seen in Section 3.7, the number of clusters does not affect the accuracy of the statistical model. However, clustering could still affect the results of the sizing problem by altering when peaks occur. In our case study, we consider only one cluster with evening peaks for comparison against the CIGRE profile with similar characteristics. However, clustering could improve the reliability of sizing results by more accurately modeling the occurrence of individual nodes' peaks. Finally, we should note that our method is not limited to using GMM as the statistical model to fit the residuals. More sophisticated models will be considered in the future, which could improve the accuracy of statistical metrics such as the correlation matrix.

5 Conclusion

Motivated by the broader challenge of distribution grid analysis and planning under electrification and energy transition, and in particular by the problem of BESSs as non-wire alternatives to grid reinforcement, we have proposed a data-driven statistical model to generate synthetic daily household electricity demand scenarios for a given distribution grid. The method relies solely on historical smart-meter measurements from multiple households, readily available in open datasets, and requires no explicit modeling of user behavior or appliances. The households are first grouped into clusters based on their consumption patterns, and then a stochastic model is fitted to the residuals for each cluster using a subset of the available measurements to reproduce demand across different days of the week or periods of the year. We show that a multivariate Gaussian mixture model (GMM) fitted on the logarithm of the residuals accurately estimates the average and median demand profiles of the original dataset, provided that extreme values are clipped. Our analysis also demonstrates that the model captures the joint distribution of daily peak power and energy consumption, which is critical not only for BESS sizing applications but, more generally, for any power-flow or hosting-capacity study in which the relation between peak demand and total energy consumption must be preserved.

We finally formulate an optimization problem to determine the optimal location and size of a BESS in a distribution network that respects the grid's operational constraints while minimizing investment costs. Comparing the sizing results obtained with synthetic scenarios generated by our method against average demand profiles reported in the literature, we show that accurately modeling the peak-power-to-energy ratio allows the sizing algorithm to compute 3.5 times less energy capacity for the considered use case, requiring 40% less investment to meet the same power requirements. These results highlight the value of the proposed method for distribution system operators (DSOs) considering the deployment of energy storage to support hosting capacity and mitigate the intermittency of distributed stochastic energy resources. Future work will focus on modeling the spatial correlation of demand profiles across network nodes and evaluating whether accounting for this correlation can improve the reliability of sizing results.

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